

Overview

You will apply text analytics to a structured business dataset to determine:

1. Whether review sentiment aligns with numeric ratings
2. Whether sentiment varies by hotel tier (cost proxy)
3. Whether there are meaningful inconsistencies between written tone and assigned score

Directions

Part 1 – Generate a Sentiment Score

Step 1: Create a Sentiment Variable

Using the reviews.text column, generate a sentiment score for each review using the following scale:

- **-1.0** = Very Negative
- **0.0** = Neutral
- **+1.0** = Very Positive

You may:

- Score each review individually
- Batch score in chunks
- Use a spreadsheet + AI workflow

But you must ensure:

- Output is numeric only
- Scores are consistently scaled
- No additional commentary is included in your sentiment column

Required Prompt Structure

Use a structured prompt similar to:

"You are performing sentiment analysis.

Assign a sentiment score between -1 (very negative) and +1 (very positive).

Return only the numeric score.

You may modify wording, but scale must remain consistent."

Part 2 – Analyze Alignment

After generating Sentiment_Score, build visualizations that explore:

1. **Sentiment vs Rating**
 - Scatterplot or aggregated comparison
 - Do higher ratings correspond to higher sentiment?
2. **Sentiment by Brand_Tier**
 - Are luxury reviews more positive?
 - Are midscale reviews more neutral?
3. **Sentiment vs Review Length**
 - Do longer reviews skew negative?
 - Are extreme ratings more verbose?

Part 3 – Identify Incongruity

Identify at least **3 reviews** where:

- Sentiment score and numeric rating strongly disagree

(e.g., 4–5 star rating but negative sentiment, or 1–2 star rating with positive tone)

For each example:

- Provide rating
- Provide sentiment score
- Provide short explanation of why the mismatch may exist

Examples of causes:

- Sarcasm
- Mixed tone
- Strong praise with one critical issue
- Mild wording with low rating

Part 4 – Interpretation (Short Written Response)

In 1–2 paragraphs, answer:

1. Does sentiment analysis meaningfully align with numeric ratings in this dataset?
2. Does tier (cost proxy) appear to influence tone?
3. What are the limitations of automated sentiment scoring in this context?

What to Include

- Submit a powerpoint deck that includes your findings.
- It should include:
 - 3 visualizations
 - 3 incongruity examples (brief explanations)
 - 1 short written interpretation
 - The exact prompt you used for sentiment generation in appendices
- **Important Reminder**
 Sentiment analysis measures tone, not truth.
 A model detecting “negative language” does not prove the hotel is low quality.
 What someone puts in the review may not reflect what really happened, just their opinion of what transpired. There is a huge difference.
 Your task is interpretation, not automation.